Number of entries	Decorate-sort-undecorate	Original version
500	0.003 seconds	1.575 seconds
3,000	0.024 seconds	4.866 seconds
150,000	0.116 seconds	18.138 seconds

Decorate-sort-undecorate is a useful technique. Eliminating repeated computation of the same thing makes the array sorting perform fast enough to be acceptable without making the program freeze for a long time. It’s just barely avoiding the problem, but sometimes avoiding the problem is exactly the right approach.

You can’t always avoid the problem, though. CoffeeScript programs run on an event loop, and an event loop does only one thing at a time.

9.2. Event loops

CoffeeScript programs, whether on Node.js or in a web browser, are executed on a single-threaded event loop where only one thing ever happens at a time. If you try to make something happen (like evaluating a CoffeeScript expression) while something else is happening, then the new thing will have to wait until the first thing is finished. To keep your CoffeeScript program from locking up, you need to adopt particular asynchronous programming techniques that are suited to a single-threaded event loop. In this section you’ll learn how to do that.

Imagine that your billboard code was so successful that Joe from Million Corporation phones you to say he wants to use it for the Intergalactic Volcano Mogul Skiing Championships. He says there are approximately 1.5 million names. A decorate-sort-undecorate version of the application with these 1.5 million names makes your program unresponsive:

```
Loading and processing data
Starting server on port 8888
Data loaded in 5.482 seconds
```